Muhammad Samir Assawalhy

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Education

Zagazig University Zagazig, Egypt
Bachelor of Electrical Engineering - Computer and Systems Engineering track *Sep 2019 – Jul 2024*

Experience

Contract Software Engineer Sep 2026 – Present
Bond Group *Riyadh, KSA (Remote)*

- Building a CRM for real estate brokers and developers as part of a two person team, with NestJS services, a Next.js front end, Drizzle ORM and PostgreSQL
- Integrated Meta lead ads through the Business account, importing Facebook and Instagram campaign leads into the CRM
- Built WhatsApp bots for lead capture and follow-up, plus Google Calendar integration
- The system is in production and has processed around 1,000 leads

Backend Intern Jun 2026 – Sep 2026
Taager *Cairo, Egypt*

- Designed and maintained microservices communicating via Apache Pulsar, in Node.js and Spring Boot Kotlin across PostgreSQL and MongoDB
- Applied Clean Architecture, DDD and CQRS, with roughly 80% unit coverage using Testcontainers and MockK
- Worked with CI/CD (ArgoCD), Kubernetes and Datadog, and used feature flags for progressive rollout and A/B experiments

Military Service (Conscription) 2024 – 2026
Egyptian Armed Forces *Egypt*

Frontend Engineer (part-time during BSc) Aug 2021 – May 2024
SciCave *Cairo, Egypt*

- Built web applications with TypeScript, React.js, Next.js and Tailwind CSS
- Tested the frontend with Cypress and automated quality checks in CI/CD pipelines
- Managed deployment with Docker and AWS EC2, and designed UI/UX in Figma while gathering requirements

Projects

TDCO Interactive Map | *Leaflet, MapLibre GL, Vite, Vanilla JS* | live Aug 2026

- Bilingual (EN/AR, RTL) map of TDCO's construction projects across all 13 Saudi regions, with POI layers, search and shareable deep links
- Built a data pipeline sourcing points of interest from OpenStreetMap

Arabi.js | *JavaScript, PEG.js* Oct 2020 – May 2021

- JavaScript Arabization based on Babel.js parser

Rasm | *JavaScript, HTML, Sass, Pugjs, PEG.js, Webpack, Gulp, Jest* Feb 2020 – Jan 2021

- Graphing calculator to plot math functions, draw geometric shapes, and animate them

Achievements

ECPC 1st place among university teams three years running; **ACPC Finalist** 2022, 2023, 2024

NASA Space Apps Challenge (Nov 2023): 3rd place and a global nominee, with team Pyrogurarians

IEEEXtreme 17.0 (Oct 2023): top 15 teams out of 198 in Egypt

Extra-Curricular Activities

Mentor and Instructor Mar 2022 – Aug 2024
ICPC Community at Zagazig University

- Mentored teams in algorithms and problem solving, delivered weekly sessions on greedy algorithms and the C++ STL, and authored problems used in local and regional ICPC contests

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, Kotlin, C++, SQL, Bash
Frontend: React.js, Next.js, Tailwind CSS, Cypress, HTML, CSS
Backend: Node.js, NestJS, Spring Boot, REST APIs, WebSockets, Apache Pulsar, PostgreSQL, MongoDB
Tools: Git, Docker, Kubernetes, Linux, AWS, CI/CD (ArgoCD), Datadog, Figma
Others: Clean Architecture, DDD, CQRS, OOP, API Design, Problem Solving